ClipSAT : Differential Equations First/Second Order, Laplace, Systems & S

Unit 9.01: Numerical Methods: Euler, RK2, RK4 & Stability

Name: _____

Q1. The second-order linear homogeneous ODE f''x - 39f'x + 350fx = 0 has characteristic roots lambda_1, lambda_2 . Determine the trace sum lambda_1 + lambda_2 .

[ ] A: 39
[ ] B: 24
[ ] C: 42
[ ] D: 53

Q2. Consider the linear first-order differential equation frac{dy}{dx} + 2y = 6 . If the initial condition is y0 = 1 , find the value of y1 .

[ ] A: 2
[ ] B: e^2 - 1
[ ] C: 3 - 2e^{-2}
[ ] D: 3 - 2e^{-1}

Q3. Consider the differential equation dfrac{dy}{dx} = x + y with the initial condition y0 = 1 . Use the Euler method with a step size of h = 0.1 to approximate y0.2 .

[ ] A: 1.200
[ ] B: 1.210
[ ] C: 1.230
[ ] D: 1.220

Q4. Consider the second-order linear homogeneous differential equation y'' - 5y' + 6y = 0 . If the auxiliary equation has roots r_1 and r_2 , find the value of r_1 + r_2 .

[ ] A: 5
[ ] B: -5
[ ] C: 6
[ ] D: -6

Q5. Consider the separable differential equation dfrac{dy}{dx} = 2x with the initial condition y0 = 3 . Find the value of y2 .

[ ] A: 7
[ ] B: 6
[ ] C: 8
[ ] D: 5

Q6. Consider the differential equation y'' - xy = 0 . The point x=0 is an ordinary point. If we assume a power series solution of the form yx = sum_{n=0}^{infty} c_n x^n , find the recurrence relation for the coefficients c_n .

[ ] A: c_{n+2} = frac{c_{n-1}}{ n+2n+1 } fo
[ ] B: c_{n+2} = frac{c_n}{ n+2n+1 } for n
[ ] C: c_{n+2} = -frac{c_{n-1}}{ n+2n+1 } f
[ ] D: c_{n+2} = -frac{c_n}{ n+2n+1 } for n

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Q7. The second-order linear homogeneous ODE u''s - 40u's + 144us = 0 has characteristic roots lambda_1, lambda_2 . Determine the trace sum lambda_1 + lambda_2 .

[ ] A: 44
[ ] B: 43
[ ] C: 24
[ ] D: 40

Q8. The Laplace transform of us = 170e^{19s} is vs = dfrac{170}{s - 19} for s > 19 . Determine the evaluated transform v20 .

[ ] A: 170
[ ] B: 151
[ ] C: 173
[ ] D: 187